

Digital Communication Design Using MATLAB



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Digital communication has been a boom for the telecommunication industry which demands higher and higher speeds with passage of time. It gives to the customer a high performance, high speed and high degree of security, while to the service provider it offers end flexibility in use, lesser need of repair and maintenance, and isolation from other service providers.

This project gives an insight to how actually digital communication works.

It gives us an idea how we can design and implement a digital communication system from the software point of view, and its advantages. The entire project has been implemented in MATLAB, which is an important tool for telecommunication engineers.

According to Wikipedia, “Data transmission, digital transmission, or digital communication is the physical transfer of data (a digital bitstream) over a point-to-point or point-to-multipoint communication channel. Examples of such channels are copper wires, optical fibres, wireless communication

tion channels and storage media. The data are represented as an electromagnetic signal, such as an electrical voltage, radiowave, microwave or infrared signal."

As we know, noise is a major factor of concern in any communication system. In early days, analogue communication was widely used, and even today many applications still rely on this mode. But analogue communication largely suffers from many types of noises as majority of information are amplitude modulated, and we know that noise affects amplitude to

its peak. With frequency modulation the noise level gets reduced though not eliminated altogether.

Digital communication uses 1's and 0's to represent the signals and then transmit them. The 1's and 0's are recovered at the receiver end by filtering out all the unwanted signals. This methodology highly reduces the noise content. The information is transmitted with amplitude-, frequency- or phase-modulation, as discussed below.

This article deals with the physical layer, i.e., designing of the transmitter and receiver using various schemes as follows:

1. Amplitude shift keying (ASK)
2. Frequency shift keying (FSK)
3. Phase shift keying (PSK)

Amplitude shift keying. In ASK scheme (refer Fig. 1), the 1's and 0's

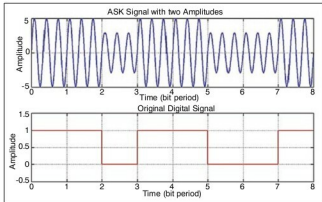


Fig. 1: ASK scheme

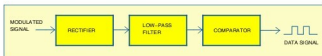


Fig. 2: Demodulation of ASK signal

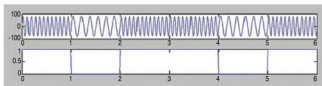


Fig. 3: FSK scheme

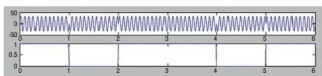


Fig. 4: PSK scheme

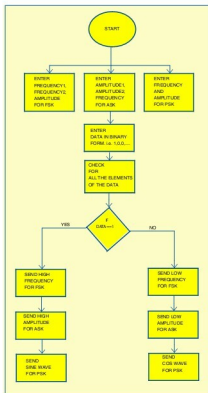


Fig. 5: Flowchart